

Lucas de Souza Lima

PYTHON DEVELOPER | DATA SCIENCE & MACHINE LEARNING | CYBERSECURITY

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PROFESSIONAL PROFILE

Quantitative technology professional and PhD researcher in Mathematics combining rigorous analytical reasoning with hands-on work in Python development, machine learning, time-series forecasting, and cybersecurity. Builds reproducible data pipelines and security tooling, evaluates predictive models with temporal validation, and documents web and network security investigations. eJPT and ICCA certified, with international research experience and professional English communication.

TECHNICAL COMPETENCIES

- Software Development:** Python, Flask, REST APIs, CLI tools, SQL, Bash, Git, Linux, automated testing
- Data Science & ML:** pandas, NumPy, scikit-learn, XGBoost, CatBoost, LightGBM, TensorFlow, supervised and unsupervised learning
- Time Series & Statistics:** ARIMA, AutoARIMA, VAR, statistical testing, feature engineering, temporal cross-validation, backtesting
- Cybersecurity:** Penetration testing, vulnerability assessment, OWASP Top 10, web and network reconnaissance, traffic analysis, Linux privilege escalation
- Security Tooling:** Nmap, Burp Suite, Wireshark, Metasploit, SQLmap, Gobuster, Hydra, John the Ripper, Hashcat
- Quantitative Foundation:** Mathematical modeling, linear algebra, calculus, differential equations, real and complex analysis

SELECTED TECHNICAL PROJECTS

- Stock ML Lab | Data Science / Software** 2026 - Present
- Designed a reproducible research framework for next-session financial-return prediction across Brazilian and US assets.
 - Implemented temporal cross-validation, classical and machine-learning baselines, hyperparameter optimization, bootstrap inference, transaction-cost analysis, and cross-asset robustness checks.
 - Compared Zero Return, Persistence, Ridge, Random Forest, XGBoost, CatBoost, ARIMA, AutoARIMA, and VAR under leakage-aware evaluation.
- AWS Cloud Cost Analysis | Data Science** 2026
- Built a data-processing and exploratory-analysis workflow for AWS instance-pricing data and trained regression models to estimate on-demand infrastructure costs.
 - Documented preprocessing, model evaluation, and business-oriented interpretation in a reproducible repository.
- WebRecon & NetRecon | Python / Cybersecurity** 2025 - 2026
- Developed Python reconnaissance tools for HTTP header inspection, endpoint discovery, CIDR host discovery, concurrent TCP scanning, banner grabbing, and service enumeration.
 - Applied threading, sockets, CLI design, and structured JSON reporting to security workflows in controlled environments.
- Bitcoin Return Forecasting | MBA Capstone** 2024 - 2025
- Evaluated linear regression, XGBoost, LightGBM, LSTM, AutoARIMA, VAR, and hybrid models on public Bitcoin data from 2014 to 2025.
 - Used rolling-window validation with MAE, RMSE, and directional accuracy, reporting near-random predictive performance transparently rather than overstating results.
- Additional work:** gold time-series analysis, diabetes-risk classification, Flask application development, and documented Hack The Box / TryHackMe case studies.

SELECTED SECURITY CASE STUDIES

- Silentium | Hack The Box:** VHost fuzzing, Flowise AI exploitation, Docker container pivoting, and Gogs remote code execution.
- Pickle Rick | TryHackMe:** Web enumeration, credential discovery, command injection, reverse-shell access, and Linux privilege escalation.
- Bounty Hacker | TryHackMe:** FTP enumeration, credential analysis, SSH access, and Linux privilege escalation using evidence-led methodology.

RESEARCH & TECHNICAL EXPERIENCE

PhD Researcher in Mathematics University of São Paulo (USP) <i>Brazil</i>	2023 - Present
- Conduct advanced research in global solvability of differential operators and systems, requiring formal analysis, abstraction, and independent problem-solving. - Structure complex technical problems into verifiable arguments and communicate results through papers, seminars, and academic presentations. - Translate rigorous mathematical reasoning into quantitative modeling and computational experimentation workflows.	
International Research Program - PDSE/CAPES University of Turin <i>Italy</i>	2025 - Present
- Collaborate in an international research environment and communicate technical results professionally in English. - Apply rigorous methodology to mathematical analysis, research documentation, and the interpretation of complex results.	
Teaching Assistant - Calculus II and III ICMC, University of São Paulo <i>Brazil</i>	2024 - 2025
- Explained complex quantitative concepts clearly and supported students in analytical and numerical problem-solving. - Planned learning activities and adapted technical communication to audiences with different levels of expertise.	
Teaching Assistant - Calculus I and Real Analysis UNESP <i>Brazil</i>	2019 - 2022
Supported students in rigorous mathematical reasoning, proof development, and structured problem-solving.	
Practical Security Labs & Research TryHackMe, Hack The Box, eJPT <i>Controlled environments</i>	2025 - 2026
- Performed web and network reconnaissance, service enumeration, exploitation, post-exploitation, traffic analysis, and Linux privilege escalation. - Investigated SQL injection, command injection, remote code execution, authentication weaknesses, exposed credentials, and system misconfigurations. - Documented technical evidence, attack paths, severity, and remediation guidance; published case studies and automated repetitive reconnaissance tasks with Python and Bash.	

EDUCATION

PhD in Mathematics University of São Paulo (USP)	2023 - Present
MBA in Finance and Controllershship ESALQ-USP	2024 - 2025
MSc in Mathematics São Paulo State University (UNESP)	2021 - 2023
BSc in Mathematics São Paulo State University (UNESP)	2018 - 2022

CERTIFICATIONS & ADDITIONAL TRAINING

eJPT - eLearnSecurity Junior Penetration Tester INE	2026
ICCA - INE Certified Cloud Associate INE	2026
Data Science with Python Santander	2025
Python AI Backend Developer DIO	2024
Complex Networks with Python and AI/LLMs USP	2024

LANGUAGES

Portuguese: Native | English: Professional proficiency